1.Implementation of Data Preprocessing

Aim :

To implement **data preprocessing** techniques that convert raw, inconsistent data into a clean and suitable form for machine learning, by performing **data loading, handling missing values, removing duplicates, outlier treatment, encoding categorical variables, feature scaling** and **train-test splitting**, and to analyze the effect of preprocessing on the data quality and **model performance**.

Algorithm :

A. Data Preprocessing Implementation

Input: Raw dataset **D** (rows = samples, columns = features and target)

Step 1: Load the dataset (for example, from a CSV file) and display its **shape, data types, first rows** and **summary statistics**.

Step 2: Handle missing values:

1. Detect the missing values in each column.
2. For **numerical** columns, fill with the **mean or median**.
3. For **categorical** columns, fill with the **mode**.
4. Drop a row or column if the proportion of missing values is too high.

Step 3: Remove duplicate records and correct inconsistent entries (for example, spelling or case differences).

Step 4: Detect and treat outliers using the **IQR method**:

1. Compute **Q1, Q3** and **IQR = Q3 − Q1**.
2. Mark values outside **[Q1 − 1.5 × IQR, Q3 + 1.5 × IQR]** as outliers.
3. **Remove or cap** (clip) these values.

Step 5: Encode categorical variables:

1. Use **Label Encoding** for ordinal features and the target.
2. Use **One-Hot Encoding** for nominal features.

Step 6: Separate the features **X** and the target **y**, then **split** the data into **training and test sets** (for example, 80:20).

Step 7: Feature scaling (fit on the training set only, then apply to both sets):

1. **Standardization: $z = (x - \mu) / \sigma$**
2. **Min-Max normalization: $x' = (x - x_{min}) / (x_{max} - x_{min})$**

Step 8: Output the cleaned and scaled training and test sets.

B. Performance Analysis Procedure

Step 1: Display the dataset **before and after** preprocessing (shape, missing-value count, duplicates).

Step 2: Compare the **feature distributions** before and after scaling using **histograms or**

box plots.

Step 3: Train a model (for example, **Logistic Regression** or **KNN**) on the **raw data** (baseline) and on the **preprocessed data**.

Step 4: Compute **accuracy, precision, recall and F1-score** for both models.

Step 5: Compare the results and **conclude** how preprocessing improved data quality and model performance.

Program&Result:

